

Pivot Table – Chapter 9 Assignment - Relationships

1. Sample Data Tables

Students Table

StudentID	StudentName	Gender	City
S001	Aarav Sharma	M	Pune
S002	Diya Patel	F	Mumbai
S003	Rohan Verma	M	Delhi
S004	Sneha Iyer	F	Chennai
S005	Arjun Mehta	M	Bangalore
S006	Kavya Nair	F	Kochi

Courses Table

CourseID	CourseName	Category
C101	Mathematics	Science
C102	Physics	Science
C103	English	Language
C104	History	Humanities
C105	Computer Science	Technology

Marks Table

MarkID	StudentID	CourseID	Marks	ExamDate
M001	S001	C101	85	2025-01-10
M002	S001	C102	78	2025-01-12
M003	S002	C101	92	2025-01-10
M004	S002	C103	88	2025-01-15
M005	S003	C104	74	2025-01-18
M006	S003	C102	81	2025-01-12
M007	S004	C105	95	2025-01-20
M008	S004	C103	89	2025-01-15
M009	S005	C101	67	2025-01-10
M010	S005	C105	72	2025-01-20
M011	S006	C104	91	2025-01-18
M012	S006	C103	87	2025-01-15

2. Relationships to Create (VERY IMPORTANT)

In Excel (Data Model):

- Students[StudentID] → Marks[StudentID]
- Courses[CourseID] → Marks[CourseID]

👉 Marks is your **fact table**

👉 Students & Courses are **dimension tables**

3. Practice Questions + Step-by-Step Solutions

✅ Q1: Total Marks per Student

Goal:

Find total marks scored by each student.

 Insight:

Who is the top performer?

 Q2: Average Marks per Course

 Goal:

Which course has highest average marks?

 Q3: Marks by Gender

 Goal:

Compare performance of male vs female students.

 Q4: Student-wise Course Performance

 Goal:

Matrix showing marks per student per course

 Q5: Highest Scoring Student in Each Course

 Goal:

Find max marks per course

 Q6: Total Marks by Category

 Goal:

Performance by subject category

 Q7: City-wise Student Performance

 Goal:

Compare cities based on average marks

✅ Q8: Monthly Performance Trend

🎯 Goal:

Analyze marks over time

✅ Q9: Top 3 Students by Total Marks

🎯 Goal:

Ranking analysis

✅ Q10: Count of Students per Course

🎯 Goal:

Enrollment analysis
